

Ilsa B. Ahmed

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SUMMARY

Software engineer who builds Python services, APIs, and data pipelines, and evaluates LLM systems against baselines instead of trusting their output. Built a multi-agent LLM pipeline whose structured prompting cut hallucinations by 52%, and traced 25+ bugs through unfamiliar production code. B.S. in Computer Science and Linguistics; CompTIA Security+ certified.

EXPERIENCE

Lawly — Software Engineer Intern

Jun 2026 – Present

- Resolved 25+ bugs in a production Next.js / React 19 and Fastify + Node.js app on PostgreSQL, partnering with QA testers to reproduce issues, trace behavior through unfamiliar code, and verify fixes before release.
- Refactored a large body of AI-generated TypeScript into readable, maintainable modules.
- Work with API integrations including WorkOS (OAuth, JWT sessions), Stripe, AWS S3, and Twilio.

Illini Solar Car — Software Engineer, Telemetry Team

Aug 2025 – Dec 2025

- Built a Python asyncio pipeline ingesting 50,000+ CAN Bus sensor data points per minute into QuestDB, designing indexing and partitioning to reduce query latency.
- Designed a FastAPI REST layer and React dashboard the team used to monitor telemetry and catch anomalies in real time.
- Containerized services with Docker and built GitHub Actions CI/CD for automated testing, linting, and image builds.

Health Care Engineering Systems Center — Data Annotator

Aug 2025 – Present

- Label eye-gaze patterns in gaze-tracking data from children with developmental disorders for research and model training.

Treevah — Software Engineer Intern

May 2025 – Aug 2025

- Integrated the Microsoft Graph API to sync nested OneDrive hierarchies, and implemented Microsoft sign-in with MSAL (OAuth 2.0) against an Azure backend.
- Co-built the core 3-panel file manager for an early-stage productivity startup (Node.js, React).

Forward Data Lab, UIUC — Undergraduate Researcher, Data Engineering

Jan 2023 – Apr 2023

- Built document-ranking pipelines with TF-IDF and vector similarity, evaluated them with precision, recall, F1, and cosine similarity, and served results through Flask REST APIs backed by PostgreSQL and MongoDB.

PROJECTS

AutoWiki — Multi-Agent Article Generator | Python, LangGraph, LangChain, Chroma

2026

- Implemented Stanford's STORM method from the paper as a LangGraph multi-agent pipeline: persona agents interview a web-searching agent, store findings in a Chroma vector database, and draft each section from that research.
- Separated retrieval, orchestration, and evaluation layers; measured output with ROUGE and BLEU, and structured prompt engineering cut the hallucination rate by 52% vs. baseline.

Gendered Discourse Markers with BERT | Python, Hugging Face Transformers, scikit-learn

- Fine-tuned BERT on the Spoken BNC 2014 conversation corpus to predict speaker gender from discourse markers like "you know," compared it with out-of-the-box BERT and a logistic regression baseline, and analyzed each model's errors.

Stock Bot Battle — Multiplayer Trading Game | C#, ASP.NET Core, PostgreSQL, React

2026 – Present

- Building a trading game where players write Python bots in a browser editor and compete on a simulated market, with transactional order execution over an append-only ledger.
- Built magic-link sign-in that stores only hashed, single-use tokens, and an audit log of sign-ins and submissions.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C#, Java

AI & LLMs: LangGraph, LangChain, RAG, Chroma, OpenAI API, Hugging Face, LLM evaluation (ROUGE, BLEU)

Data & ML: PostgreSQL, MongoDB, QuestDB, pandas, NumPy, scikit-learn, spaCy, PyTorch

Back End & Front End: FastAPI, Flask, Node.js, Fastify, ASP.NET Core, REST APIs, React, Next.js

Infra & Tools: Docker, GitHub Actions CI/CD, AWS (EC2, S3), Azure, Twilio, Git

EDUCATION

University of Illinois Urbana-Champaign — B.S. Computer Science and Linguistics

May 2026

Certifications: CompTIA Security+ (2026); Aspiritech Academy Cybersecurity Trainee, Security+ & CMMC (Jun 2026)